

KIYARI VOICE ASSISTANT - OFFLINE OPERATION GUIDE

Welcome to KIYARI!

This guide contains everything you need to operate the KIYARI Voice Assistant completely offline tomorrow. Follow these steps to boot, connect, and utilize the assistant without any internet connection.

1. QUICK START STEPS (WHAT TO DO TOMORROW)

1. Turn on your local LLM Server (LM Studio) on your PC. Make sure the model 'qwen2.5-7b-instruct' is loaded.
2. Connect your PC to the Wi-Fi network 'lucifer'.
3. Open your PC terminal (PowerShell), navigate to the backend folder:
`cd c:\Users\VISHAV\OneDrive\Desktop\KIYARI\backend`
4. Run the backend server using the virtual environment:
`.\env\Scripts\python.exe server.py`
5. Plug in the KIYARI ESP32-S3 speaker hardware to power. It will automatically connect to 'lucifer' Wi-Fi and establish a WebSocket connection to the server on port 8001.
6. Once connected, the ESP32 OLED screen will display:
Connected
Ready
Press BOOT to speak

2. HOW TO OPERATE THE VOICE ASSISTANT

* SPEAKING TO KIYARI:

- Press and hold the BOOT button (GPIO 0) on the ESP32-S3 board.
- Speak your command clearly into the microphone (you can speak in English, Hindi, or bilingual Hinglish).
- Release the BOOT button the moment you finish speaking.
- The assistant will show 'Thinking...' on the screen, process your query, and then play back the response dynamically in a natural Indian English/Hindi voice!

* ADJUSTING VOLUME BY VOICE:

- Speak to the assistant to change the volume directly! Examples:
 - 'Set volume to 50%'
 - 'Increase volume' or 'Louder'
 - 'Decrease volume' or 'Quieter'
- The backend will adjust the volume digitally, apply clipping protection to keep the audio clear, and respond back confirming the change.

3. BILINGUAL SPEECH & TTS VOICE MODELS

- KIYARI supports both English and Hindi voice responses.
- Whisper automatically transcribes what you speak, and the LLM responds in the appropriate language.
- If the response is in Hindi/Hinglish, KIYARI will speak using a native Hindi male voice (hi_IN-pratham-medium).
- If the response is in English, KIYARI will speak using a clear English female voice (en_US-lessac-medium).

4. TROUBLESHOOTING (OFFLINE CHECKS)

- SCREEN SHOWS 'Error occurred':
 - Press the RST button on the ESP32 board to reboot it, or unplug/replug the USB cable. It will reconnect automatically.
- PORT 8001 ALREADY IN USE:

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- If the server fails to start, kill any active process on port 8001 by running this in PowerShell:
Stop-Process -Id (Get-NetTCPConnection -LocalPort 8001).OwningProcess -Force
- NO VOICE RESPONSE:
 - Make sure LM Studio is running on your PC at <http://127.0.0.1:1234> and the LLM is loaded.